

1. PROTOTYPE TECHNOLOGIES & OPERATIONAL METHODOLOGY

5-TIER END-TO-END PIPELINE · SUB-10MS INFERENCE · ZERO-HALLUCINATION

01. SOURCING

- **39 Defense Outposts:**
Siachen, DBO, Dras, Galwan
- **NASA POWER & ISHRAE:**
Diurnal subzero climate (-40°C)
- **CPWD DSR 2023 DB:**
102 envelope materials (k, rho)

02. DETERMINISTIC CORE

- **1D RC Heat Solver:**
EN ISO 52016-1 multi-node
- **Barometric Scaling:**
Altitude density (0.81 kg/m³)
- **Swinbank Sky Sink:**
Clear-sky nocturnal longwave

03. SCIENTIFIC ML

- **120k Dataset:**
Physics-grounded master rows
- **5 ML Surrogates:**
HistGBDT, RF, MLP (R²=0.968)
- **Sub-10ms Inference:**
40,000x faster than EnergyPlus

04. MULTI-OBJECTIVE MCDA

- **Pareto Frontier:**
Strict cost vs discomfort trade-off
- **Utopia Knee Point:**
Normalized Euclidean distance Di
- **Safety Interlock:**
Combustion + ACH < 0.35 => REFUSED

05. OPERATIONAL C2

- **FastAPI + React 18/Vite:**
Frozen REST JSON contract
- **Three.js 3D WebGL:**
Interactive shelter peel cutaway
- **Voice Diagnostic Orb:**
Hands-free verbal field inquiry

A. SOLAR GEOMETRY & HIGH-ALTITUDE IRRADIANCE

Duffie & Beckman / Perez Solar Model

Solar Declination Angle (delta):

$$\delta = 23.45^\circ \times \sin[(360^\circ / 365) \times (284 + n)]$$

↳ n = day of year [1..365]; orbital obliquity tilt calculation

Solar Altitude Angle (alpha_s):

$$\sin \alpha_s = \sin \phi \sin \delta + \cos \phi \cos \delta \cos \omega$$

↳ φ = latitude, ω = 15° (t_{solar} - 12) hour angle

Surface Incidence Angle Clamp (cos theta):

$$\cos \theta = \max(\sin \alpha_s \cos \beta + \cos \alpha_s \sin \beta \cos(\gamma_s - \gamma_{surf}), 0.0)$$

↳ Clamped at zero: prevents non-physical negative solar cooling

Total Absorbed Surface Solar Irradiance:

$$I_{tot} = I_{beam} \cos \theta + I_{diff} \frac{1 + \cos \beta}{2} + GHI \cdot \rho_{snow} \frac{1 - \cos \beta}{2}$$

↳ Direct beam + isotropic diffuse + snow reflected (ρ_{snow} = 0.75)

C. RAREFIED ATMOSPHERE, SKY SINK & INFILTRATION

ISA Barometric Model / Swinbank Radiant Sink

High-Altitude Atmospheric Pressure:

$$P(h) = 101325 \cdot (1 - 2.25577 \times 10^{-5} h)^{5.25588} \text{ Pa}$$

↳ Barometric pressure formula at elevation h metres above sea level

Temperature-Corrected Air Density:

$$\rho_{air}(h, T) = \frac{P(h)}{287.058 \cdot T_{air}} \text{ kg/m}^3$$

↳ At Leh (3,500m), air density drops to ≈ 0.81 kg/m³ (35% lower convective loss)

Sensible Infiltration Heat Dissipation:

$$Q_{inf} = \frac{ACH \times V \times \rho_{air} \times 1005 \times (T_{in} - T_{out})}{3600} \text{ W}$$

↳ C_{p,air} = 1005 J/(kg·K); scales with altitude-reduced density ρ_{air}

Swinbank Nocturnal Clear-Sky Radiation Sink:

$$T_{sky} = 0.0552 \cdot T_{air}^{1.5} \text{ K}, \quad Q_{sky} = h_r A F_{sky} (T_s - T_{sky})$$

↳ σ = 5.67 × 10⁻⁸, F_{sky} = 1.0 (roof), 0.5 (walls); severe subzero sink

B. TRANSIENT RC FOURIER SOLVER & STABILITY

EN ISO 52016-1 / 1D Multi-Layer RC Discretization

1D Heat Diffusion Equation:

$$\rho C_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} (k \frac{\partial T}{\partial x})$$

↳ Dynamic transient heat transfer through multi-layer envelope

Nodal Capacitance & Conductance:

$$C_i = \rho C_p \Delta x A, \quad K_{i,i+1} = \frac{kA}{\Delta x}$$

↳ Discretised nodal thermal mass (J/K) and conductances (W/K)

Explicit Euler Stability Ceiling (Fourier Number):

$$Fo = \frac{\alpha \Delta t}{(\Delta x)^2} \leq 0.25 \Rightarrow \Delta x_{max} = \sqrt{\frac{\alpha \Delta t}{Fo_{target}}}$$

↳ α = k/(ρC_p) diffusivity; dynamically prevents numerical divergence

Assembly Thermal Resistance & Overall U-Value:

$$R_{tot} = R_{si} + \sum_{k,j} R_{kj} + R_{se}, \quad U = \frac{1}{R_{tot}}$$

↳ ISO 6946 film resistances: R_{si} = 0.13, R_{se} = 0.04 m²K/W

D. THERMAL ATTRIBUTION, PARETO MCDA & SAFETY

DRDO Decision Framework / ASHRAE 62.2 Safety

100% Component Heat Loss Attribution & Bottleneck:

$$P_k = \frac{Q_k}{\sum Q_{loss}} \times 100\%, \quad k^* = \operatorname{argmax}_k (P_k)$$

↳ Identifies primary thermodynamic loss vector (Sky vs Wall vs Infiltration)

Pareto Dominance Principle (Cost vs Discomfort):

$$A \text{ dominates } B: \text{Cost}_A \leq \text{Cost}_B \text{ and } \text{Discomf}_A \leq \text{Discomf}_B$$

↳ Strict multi-objective minimization without arbitrary subjective weighting

Normalized Utopia Distance (Balanced Pareto Knee):

$$D_i = \sqrt{(1 - c_i^*)^2 + (k_i^* - 0)^2} \rightarrow \min$$

↳ Euclidean distance to ideal point (Comfort=1.0, Cost=0.0) selects balanced retrofit

Deterministic Combustion Safety Interlock:

$$\text{Combustion Heater and } ACH < 0.35 \rightarrow \text{REFUSED}$$

↳ Zero-hallucination life safety interlock prevents fatal Carbon Monoxide asphyxiation